

MedAssist AI – Synthetic Clinical Documentation Guide

Test document for an academic RAG prototype

"This document contains fictional training content created only for software testing and demonstration."

CORPUS IDENTIFIER

KB-SYNTH-MED-001

DOCUMENT TARGET

RAG Retrieval & Chunking

DOMAIN CONTEXT

Computer Science Capstone

The MedAssist AI synthetic documentation guide provides formalized data schema definitions, structured entry protocols, and syntactic requirements designed specifically for natural language retrieval and text chunking evaluation in an academic setting.

1 Blood Pressure Documentation

A blood pressure documentation entry should record the **date and time** of the reading. Systematic temporal recording ensures that retrieval systems can disambiguate successive physiologic measurements across distinct observation instances.

The patient's **position** should be documented, such as:

- **Sitting:** Standard upright posture with back supported and feet flat on the floor.
- **Standing:** Orthostatic posture recorded during vertical stability evaluation.
- **Lying down:** Supine position documented during recumbent assessment.

The **measurement site** and **cuff information** should also be included when relevant to capture accurate procedural metadata.

The entry should include:

- Systolic pressure
- Diastolic pressure
- Pulse rate
- Symptoms reported at the time of measurement
- Relevant medication context

EXAMPLE DOCUMENTATION FIELDS

Date:	[YYYY-MM-DD]	Diastolic:	[Value in mmHg]
Time:	[HH:MM 24hr]	Pulse:	[Beats per minute]
Position:	[Sitting / Standing / Lying down]	Symptoms:	[Documented patient complaints]
Measurement Site:	[e.g., Left arm / Right arm]	Medication Context:	[Active antihypertensive or relevant agent]
Cuff Information:	[Standard adult / Large adult / Thigh]	Systolic:	[Value in mmHg]

2 Medication Reconciliation

Medication reconciliation is a formal review process designed to compile an accurate and comprehensive accounting of an individual's complete pharmacological profile across transitions. Medication reconciliation should record the following information:

- **Medication name:** The distinct pharmaceutical brand name or generic designation.
- **Strength:** The specific active dosage quantity per administration unit.
- **Route:** The pathway of drug administration (e.g., oral, intravenous, transdermal).
- **Frequency:** The specified schedule and administration intervals.
- **Known allergies:** Documented hypersensitivities, previous adverse drug events, and reactions.
- **Verification status:** The confirmation indicator validating the source of information.

When a medication list is reviewed, differences between the **reported medication list** and the **available medication record** should be clearly documented. Identifying these discrepancies enables software routines to cross-reference conflicting datasets.

EXAMPLE DOCUMENTATION FIELDS

Medication Name:	<i>[Documented pharmaceutical agent]</i>	Known Allergies:	<i>[Specific allergen & reaction profile]</i>
Strength:	<i>[Dosage potency, e.g., mg or mcg]</i>	Verification Status:	<i>[Verified / Pending confirmation]</i>
Route:	<i>[Administration path]</i>	Differences Identified:	<i>[Variances between reported vs recorded]</i>
Frequency:	<i>[Timing interval schedule]</i>	Review Date:	<i>[YYYY-MM-DD reconciliation date]</i>

3 Basic Observation Record

A basic observation record provides standard numeric or categorical logging of physiological parameters. A basic observation record can include:

- **Observation date:** The calendar date on which the measurement took place.
- **Observation time:** The precise time the observation was acquired.
- **Measurement:** The numerical value or qualitative state recorded.
- **Unit of measurement:** The standardized unit (e.g., bpm, Celsius, kg, %).
- **Source of measurement:** The equipment, automated telemetry, clinician, or reported origin.
- **Reported symptoms:** Concurrent subjective sensations or concerns voiced during entry.

EXAMPLE DOCUMENTATION FIELDS

Observation Date:	<i>[YYYY-MM-DD]</i>	Unit:	<i>[Metric standard unit symbol]</i>
Observation Time:	<i>[HH:MM 24hr]</i>	Source:	<i>[Observer type / Instrument model]</i>

4 Documentation Quality

Standardized records ensure clarity, integrity, and traceability for evaluation across AI retrieval pipelines. Good documentation should be:

- **Clear:** Unambiguous language that leaves no room for multiple interpretations.
- **Dated:** Explicit temporal indicators attached to all recorded parameters.
- **Specific:** Exact figures, explicit physical locations, and distinct descriptors.
- **Easy to understand:** Straightforward syntax that simplifies computational entity parsing.
- **Recorded close to the time of the observation or review:** Minimizes recall decay and ensures contextual alignment.

Avoid unexplained abbreviations when a clearer description is available. Non-standard acronyms increase semantic drift and risk misclassification in semantic embeddings.

Important information should be recorded close to the time when the observation or review occurs. Prompt logging guarantees accurate sequence serialization in vector indexing.

5 Quick Reference Table

The following matrix summarizes the standard data schema components across the foundational clinical documentation modules specified in this test guide:

Topic	Key Information
Blood pressure	Date/time, position, cuff/site, systolic, diastolic, pulse, symptoms, medication context
Medication reconciliation	Name, strength, route, frequency, allergies, verification status
Observation record	Date/time, measurement, unit, source, reported symptoms
Documentation quality	Clear, dated, specific, minimal unexplained abbreviations

Chunking Directive for RAG Indexing: The table above is structured for exact row-wise retrieval. In vector evaluation experiments, question queries matching key schema fields should retrieve the parent topic node with high similarity confidence.

6 RAG Test Questions

The following evaluation questions are designed to benchmark the retrieval accuracy, semantic relevance, and grounded synthesis capabilities of the MedAssist AI RAG prototype. Every question is directly answerable from the factual content in this document.

1. What information should be recorded for a blood pressure reading?

Target Context: Section 1 (Blood Pressure Documentation) • Date/time, position, site/cuff, systolic, diastolic, pulse, symptoms, medication context

2. Which patient positions can be documented during a blood pressure measurement?

Target Context: Section 1 (Blood Pressure Documentation) • Sitting, standing, and lying down

3. What pulse-related information should be included in a blood pressure entry?

Target Context: Section 1 (Blood Pressure Documentation) • Pulse rate recorded in beats per minute

4. What details should be recorded during medication reconciliation?

Target Context: Section 2 (Medication Reconciliation) • Medication name, strength, route, frequency, known allergies, verification status

5. What should be documented about allergies during medication reconciliation?

Target Context: Section 2 (Medication Reconciliation) • Known allergies, previous adverse drug events, and reactions

6. What does the verification status refer to in a medication reconciliation record?

Target Context: Section 2 (Medication Reconciliation) • The confirmation indicator validating the source of information

7. What information belongs in a basic observation record?

Target Context: Section 3 (Basic Observation Record) • Observation date, time, measurement, unit, source, and reported symptoms

8. Why should documentation be clear, dated, and specific?

Target Context: Section 4 (Documentation Quality) • To ensure clarity, integrity, unambiguous meaning, and traceability across systems

9. What should be avoided when writing documentation?

Target Context: Section 4 (Documentation Quality) • Avoid unexplained abbreviations when a clearer description is available

10. When should information ideally be recorded?

Target Context: Section 4 (Documentation Quality) • Close to the time when the observation or review occurs